

July 13 (a)—© typeface by Chris Lozos

INSPIRED BY A TYPOPHILE BATTLE WEEK INVOLVING LIGATURE MADNESS WITH COOPER BLACK. FROGGY WAS MY DOUBLE “G” LIGATURE CARRIED ON TO SOMETHING ELSE, A SEMI-SERIF.

OHAMBURGEFONSTIV

**A B C D E F G H I J K L M N O P Q R S T U V
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GHIJKLMNOPQRSTUVWXYZ

AFTER WE LIVED AND REAPED OUR BOUNTY, MANY YEARS

QUIETLY Liggy bagvern gwen

Tačnije izražavanje istih. Naime naiđoh na jednu veoma interesantnu diskusiju na Typophile forumima o označavanju ironije u tekstu.

Koliko puta sam se samo osećao uskraćeno kada sam hteo misao da prenesem u pisanu formu, a na kraju ostao pitati se: “Da li će ovo neko uopšte da skapira?”. U svakom slučaju, radi se o arapskom znaku pitanja (koji je u suštini horizontalno izvrnut običan upitnik) koji bi se u engleskom (potencijalno i u drugim jezicima) koristio u svrhe označavanja ironije u tekstu.

Ovo je sada nameće pitanje da li je pametno uzimati karakter iz jednog jezika koji tamo služi jednoj svrsi, koji bi predstavljao nešto sasvim treće u nekom sasvim trećem jeziku? Da li to dovodi do dodatne konfuzije? Primer mogu biti navodnici recimo, u engleskom su oba navodnika gornja, a u srpskom, otvarajući navodnik ide dole, a zatvarajući gore. Nije li to komplikovano? Pogedajte samo Web kao primer—retko ko se bakće sa pravilnom tipografskom punktuacijom, a pogotovo se retko ko bavi pravilnom tipografskom punktuacijom u srpskom jeziku na Webu.

Ja se donekle trudim da ispoštujem tipografiju koliko mi vreme, poznavanje i sam CMS dozvoljavaju, i ujedno i apelujem na druge da pokušaju da urade isto. Gledam da primereno koristim duge i kratke srednje crte (engl. em i en dash), pravim razmak posle interpunkcijskih znakova, koristim elipsis za trotačku (da, trotačka nije tri obične tačke, već jedan karakter), pazim na dužinu pasusa, odvajam tekst u pasuse što je veoma bitno. Koliko puta sam samo dobio kilometarski elektronski email (da, i tako ga zovu) koji je napisan, što će reći u jednom dahu, tj. bez pasusa. Totalno se iznerviram kada mi se desi ovakva stvar. O pravopisu i da ne govorim.

Zanimljiv fenomen koji se nikako ne može zaobići jeste i opštepoznati i odveć omniprezentni smajli bez koga ne znam kakvo bi stanje i sporazumevanje bilo na webu.

magyar (Hungarian)

Károli Gáspár (1908)

Magyar Bibliatársulat újjordítású Bibliája (protestáns)

Szent István Társulati Biblia (katolikus)

Mind az egész földnek pedig egy nyelve és egyféle beszéde vala.

És lőn mikor kelet felől elindultak vala, Sineár földén egy síkságot találának és ott letelepedének.

És mondának egymásnak: Jertek, vessünk téglát és égessük ki jól; és lőn nekik a téglá kő gyanánt, a szurok pedig ragasztó gyanánt.

És mondának: Jertek, építsünk magunknak várost és tornyot, melynek teteje az eget érje, és szerezzünk magunknak nevet, hogy el ne széledjünk az egész földnek színén.

Az Úr pedig leszálla, hogy lássa a várost és a tornyot, melyet építenek vala az emberek fiai.

És monda az Úr: Imé e nép egy, s az egésznek egy a nyelve, és munkájának ez a kezdete; és bizony semmi sem gátolja, hogy véghez ne vigyenek mindent, a mit elgondolnak magukban.

Nosza szálljunk alá, és zavarjuk ott össze nyelvöket, hogy meg ne értsék egymás beszédét.

És elszéleszté őket onnan az Úr az egész földnek színére; és megszűnének építeni a várost.

Kekchi

Saß eb li cutan aßan jun ajcuiß li râtinobâleb li cristian saß ruchichßochß.

Li cristian queßcôeb chi cuânc saß li naßajej li cuan bar na-el cuiß chak li sakße. Queßxtau jun chßochß helho ru. Li chßochß aßan cuan xcuênt Sinar. Ut aran queßcana.

Ut queßxye chi ribileb rib: –Takayîb li xan ut takacßat chi us, chanqueb. Ut li xan quicßanjelac chokß rêkaj li pec. Ut li quicßanjelac re xletzbal li xan, aßan li na-el saß chßochß asfalto xcßababaß.

Queßxye: –Kayîbak junak katenamit ut jun li nimla ochoch najt xteram torre xcßababaß. Tâcuulak toj saß choxa li xnajtil xteram. Chi joßcaßin teßxqßue retal saß chixjunil li ruchichßochß nak lâo cuan kanaßleb. Ut incßaß takajeqßui kib yalak bar saß ruchichßochß, chanqueb.

Slovenský (Slovak)

Vtedy celý svet používal jedinú reč a rovnaké slová.

Keď sa na východe ľudia pohli, našli rovinu v krajine Šineár a usadili sa tam.

Povedali si: Poďme, narobme si tehál a dobre ich vypálme. Tak im tehly slúžili za kameň a asfalt za maltu.

Potom si povedali: Poďme, vystavme si mesto a vežu, ktorej vrch by siahahal po nebesá; tak si urobíme meno, aby sme neboli roztratení po celej zemi.

Nato zostúpil Hospodin, aby videl mesto i vežu, ktorú stavali ľudskí synovia.

Hospodin riekol: Je tu jeden ľud a všetci majú jednu reč; a to je len začiatok ich výčinov; teraz im už nič neznemožní vykonať, čo si zaumienia.

Poďme, zostúpme a zmaťme im tam reč, aby ani jeden nerozumel reči druhého.

Tak ich Hospodin rozptýlil po celej zemi a prestali stavať mesto.

Und sie sprachen einer zum anderen: Wohlan, laßt uns Ziegel streichen und hart brennen! Und der Ziegel diene ihnen als Stein, und das Erdharz diene ihnen als Mörtel.

Und sie sprachen: Wohlan, bauen wir uns eine Stadt und einen Turm, dessen Spitze an den Himmel reiche, und machen wir uns einen Namen, daß wir nicht zerstreut werden über die ganze Erde!

Und Jehova fuhr hernieder, die Stadt und den Turm zu sehen, welche die Menschenkinder bauten.

Und Jehova sprach: Siehe, sie sind ein Volk und haben alle eine Sprache, und dies haben sie angefangen zu tun; und nun wird ihnen nichts verwehrt werden, was sie zu tun ersinnen.

Wohlan, laßt uns herniederfahren und ihre Sprache daselbst verwirren, daß sie einer des anderen Sprache nicht verstehen!

Und Jehova zerstreute sie von dannen über die ganze Erde; und sie hörten auf, die Stadt zu bauen.

Darum gab man ihr den Namen Babel; (Verwirrung) denn daselbst verwirrte Jehova die Sprache der ganzen Erde, und von dannen zerstreute sie Jehova über die ganze Erde.

PRZETO ZGROMADZENIE OGÓLNE

Ogłasza Uroczyście niniejszą Powszechną Deklarację Praw Człowieka jako wspólny najwyższy cel wszystkich ludów i wszystkich narodów, aby wszyscy ludzie i wszystkie organy społeczeństwa – mając stale w pamięci niniejszą Deklarację – dążyły w drodze nauczania i wychowywania do rozwijania poszanowania tych praw i wolności i aby zapewniły za pomocą postępowych środków o zasięgu krajowym i międzynarodowym powszechnie i skuteczne uznanie i stosowanie tej Deklaracji zarówno wśród narodów Państw Członkowskich, jak i wśród narodów zamieszkujących obszary podległe ich władzy.

Artykuł 1

Wszyscy ludzie rodzą się wolni i równi pod względem swej godności i swych praw. Są oni obdarzeni rozumem i sumieniem i powinni postępować wobec innych w duchu braterstwa.

Każdy człowiek posiada wszystkie prawa i wolności zawarte w niniejszej Deklaracji bez względu na jakiekolwiek różnice rasy, koloru, płci, języka, wyznania, poglądów politycznych i innych, narodowości, pochodzenia społecznego, majątku, urodzenia lub jakiegokolwiek innego stanu.

Nie wolno ponadto czynić żadnej różnicy w zależności od sytuacji politycznej, prawnej lub międzynarodowej kraju lub obszaru, do którego dana osoba przynależy, bez względu na to, czy dany kraj lub obszar jest niepodległy, czy też podlega systemowi powiernictwa, nie rządzi się samodzielnie lub jest w jakikolwiek sposób ograniczony w swej niepodległości.

FOCUS ON DARK MATTER

KLAUS PRETZL 2000 NEW J. PHYS.
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THIS 'FOCUS ON DARK MATTER' CLUSTER OF ARTICLES IS THE VERY FIRST IN A SERIES OF FEATURED TOPICS TO APPEAR IN NEW JOURNAL OF PHYSICS THAT WILL BE OF GREAT INTEREST TO THE PHYSICS COMMUNITY. THE IDEA IS TO BRING TOGETHER CONTRIBUTIONS FROM LEADING RESEARCHERS IN TOPICAL FIELDS TO PROVIDE INSIGHTS INTO THE KEY ISSUES FOR BOTH EXPERTS AND NON-SPECIALISTS ALIKE.

INVESTIGATIONS OF DARK MATTER IN THE UNIVERSE ARE OF FUNDAMENTAL INTEREST TO ASTRONOMERS, ASTROPHYSICISTS, COSMOLOGISTS, AND NUCLEAR AND PARTICLE PHYSICISTS. RECENT DEVELOPMENTS HAVE PROVIDED NEW INSIGHT INTO THE MATTER BUDGET OF THE UNIVERSE. USING SUPERNOVAE OF TYPE 1A AS STANDARD CANDLES, DISTANCE MEASUREMENTS INDICATE AN ACCELERATING EXPANSION OF THE UNIVERSE. THIS RESULT SUGGESTS A NON-ZERO COSMOLOGICAL CONSTANT, AND THE RESULTING VACUUM ENERGY DENSITY CONTRIBUTES 70% OF THE CRITICAL MASS OF THE UNIVERSE. GRAVITATIONAL LENSING, ORIGINALLY PROPOSED BY EINSTEIN AND LATER UTILIZED FOR

MASS DETERMINATION OF GALACTIC CLUSTERS BY F ZWICKY AND OTHERS, HAS RECENTLY BEEN USED TO MAP DARK MATTER IN CLUSTERS AND LARGE-SCALE STRUCTURES. THESE EXPERIMENTS SHOW THAT 30% OF THE CRITICAL MASS IS CONTAINED IN THESE STRUCTURES, OF WHICH ONLY 3% IS VISIBLE. SINCE THE BARYONIC COMPONENT OF MATTER IS ONLY 6% OF THE CRITICAL MASS, ASSUMING A HUBBLE CONSTANT OF $60 \text{ km s}^{-1} \text{ Mpc}^{-1}$, THERE IS FIVE TIMES MORE DARK MATTER THAN BARYONIC MATTER IN THE UNIVERSE. PRIME CANDIDATES FOR MAKING UP THIS DEFICIT ARE NEUTRINOS WITH MASS (HOT DARK MATTER) AND WEAKLY INTERACTING MASSIVE PARTICLES, THE SO-CALLED WIMPS (COLD DARK MATTER). COSMOLOGISTS AND PARTICLE PHYSICISTS FAVOUR THE LATTER TO BE THE LIGHTEST SUPERSYMMETRIC PARTICLE, THE SO-CALLED NEUTRALINO, SINCE IF IT DOES EXIST IT WOULD SOLVE TWO PROBLEMS AT THE SAME TIME, NAMELY COLD DARK MATTER AND GRAND UNIFICATION THEORIES. COSMIC BACKGROUND MICROWAVE RADIATION DATA AND LARGE-SCALE STRUCTURE DISTRIBUTIONS, HOWEVER, FAVOUR A COCKTAIL OF 30% HOT AND 70% COLD DARK MATTER.

TREUMANN'S GROUP INVESTIGATES MATTER DISTRIBUTIONS IN GALACTIC CLUSTERS, INCLUDING MASSIVE NEUTRINOS AND COLLISIONAL INTRACLUSTER GAS EMITTING X-RAY RADIATION. THE ISSUE OF BARYONIC DARK MATTER IN OUR GALACTIC HALO IS EXPLORED IN THE CONTRIBUTION FROM DE PAOLIS AND COWORKERS. THEY SHOW THAT THE OBSERVED DIFFUSE GAMMA-RAY EMISSION FROM THE GALACTIC HALO CAN BE EXPLAINED BY COSMIC RAY PROTON SCATTERING ON CLOUDS OF DARK CLUSTERS OF BROWN DWARFS AND COLD SELF-GRAVITATING H_2 CLOUDS IN THE HALO. NEW EXPERIMENTAL RESULTS ON TERRESTRIAL WIMP SEARCHES VIA WIMP-Xe INELASTIC SCATTERING ARE PRESENTED BY BERNABEI'S GROUP. THE CONTRIBUTION FROM CEBRIÁN'S TEAM DESCRIBES PRESENT AND FUTURE WIMP EXPERIMENTS USING CONVENTIONAL AND CRYOGENIC DETECTORS IN THE CANFRANC UNDERGROUND LABORATORY. AN INTERESTING NOVEL LARGE-MASS, LOW-BACKGROUND, SUPERHEATED DROPLET DETECTOR FOR WIMP SEARCHES IS DESCRIBED BY COLLAR AND COLLEAGUES.

PHYSICS GETS DARK AND EXOTIC

EDITORIAL: JANUARY 2000

TWO OF THE OUTSTANDING CHALLENGES IN PHYSICS IDENTIFIED IN OUR MILLENNIUM SURVEY LAST MONTH WERE THE NATURE OF “DARK MATTER” AND A PROPER UNDERSTANDING OF NUCLEAR STRUCTURE. THIS MONTH WE LOOK AT THESE CHALLENGES IN GREATER DETAIL.

DARK MATTER IS MATTER THAT DOES NOT INTERACT WITH ELECTROMAGNETIC RADIATION: IT CANNOT BE SEEN WITH TELESCOPES AND ONLY REVEALS ITSELF THROUGH ITS GRAVITATIONAL INTERACTIONS. ASTRONOMERS FIRST BECAME AWARE IN THE 1930S THAT OUR MILKY WAY GALAXY WAS ROTATING FASTER THAN COULD BE EXPLAINED BY THE GRAVITATIONAL INFLUENCE OF THE STARS AND DUST THAT IT CONTAINED. IT LATER BECAME CLEAR THAT MORE THAN 90% OF THE MATTER IN THE UNIVERSE—AND POSSIBLY AS MUCH AS 99%—MIGHT BE DARK.

SOME OF THE DARK MATTER MIGHT BE IN THE FORM OF ORDINARY OR “BARYONIC” MATTER MADE PRIMARILY OF NEUTRONS AND PROTONS—SUCH AS FAILED STARS OR BLACK HOLES. A SMALL FRACTION MIGHT BE IN THE FORM OF NEUTRINOS, BUT THE MAJORITY OF DARK MATTER IS NON-BARYONIC AND BEYOND THE STANDARD MODEL OF PARTICLE PHYSICS. NEUTRINOS WITH MASS ARE ALSO BEYOND THE STANDARD MODEL BUT NOT AS FAR BEYOND AS NEUTRALINOS, AXIONS AND OTHER EXOTIC PARTICLES. ANOTHER POSSIBILITY FOR THIS MISSING MASS IS “DARK ENERGY” IN THE FORM OF A COSMOLOGICAL CONSTANT.

IN THIS ISSUE NIGEL SMITH AND NEIL SPOONER OF THE UK DARK MATTER COLLABORATION DESCRIBE EFFORTS TO DETECT DARK MATTER IN UNDERGROUND EXPERIMENTS. ALTHOUGH THE EXPERIMENTS SOUND EASY—CONNECT SOME PHOTOMULTIPLIER TUBES TO A CRYSTAL AND COUNT THE FLASHES—THE CHALLENGE OF ISOLATING A DARK-MATTER SIGNAL AMONGST A MYRIAD OF OTHER BACKGROUND EFFECTS IS FORMIDABLE. HOWEVER, DARK-MATTER SEARCHES CAN BE PERFORMED BY TEAMS MUCH SMALLER THAN THOSE FOUND IN TRADITIONAL PARTICLE-PHYSICS EXPERIMENTS. THE UNEQUIVOCAL DETECTION OF A DARK-MATTER PARTICLE WOULD BE A TREMENDOUS BOOST FOR PARTICLE PHYSICS AND COSMOLOGY, AL-

THOUGH THERE WOULD DOUBTLESS BE CALLS FOR IT TO BE CONFIRMED IN AN ACCELERATOR-BASED EXPERIMENT.

BACK IN THE BARYONIC WORLD, MEANWHILE, NUCLEAR PHYSICISTS ARE PROBING MORE AND MORE EXOTIC NUCLEI. OVER 7000 DIFFERENT NUCLEI ARE THOUGHT TO EXIST, BUT ONLY 3000 OF THESE HAVE BEEN CREATED AND STUDIED IN THE LABORATORY, AND ONLY 260 OF THE TOTAL ARE STABLE. AS ISAO TANIHATA EXPLAINED LAST MONTH, THERE ARE VARIOUS MODELS OF THE NUCLEUS THAT WORK WELL IN DIFFERENT REGIONS OF THE CHART OF THE NUCLIDES—SOMETHING THAT IS A SOURCE OF BOTH PRIDE AND FRUSTRATION TO NUCLEAR PHYSICISTS. IN THIS ISSUE PADDY REGAN AND BERTRAM BLANKDESCRIBE ATTEMPTS TO SYNTHESIZE AND STUDY NUCLEI AT THE VERY LIMITS OF STABILITY—THE PROTON AND NEUTRON DRIPLINES.

THESE EXOTIC SPECIES CAN BE DETECTED AMONG THE DEBRIS OF COLLISIONS BETWEEN STABLE NUCLEI. HOWEVER, IF ENOUGH OF THESE SHORT-LIVED NUCLEI COULD BE ISOLATED AND THEN RE-ACCELERATED IN A RADIOACTIVE BEAM, IT WOULD BE POSSIBLE TO CREATE AND STUDY EVEN MORE EXOTIC NUCLEI. PLANS FOR NEW FACILITIES TO PERFORM SUCH EXPERIMENTS ARE WELL ADVANCED IN VARIOUS NUCLEAR-PHYSICS LABS AROUND THE WORLD. PRETTY KITTY OTTO

AS WE EMBARK ON THE 21ST CENTURY, DARK MATTER AND NUCLEAR STRUCTURE COULD BE TWO OF THE OUTSTANDING CHALLENGES IN PHYSICS TO BE OVERCOME FIRST.

**ULTIMATE LIMITS TO
INERTIAL MASS**

SENSING BASED UPON NANOELECTROMECHANICAL SYSTEMS

AUTHORS: EKINCI, K. L.; YANG, Y. T.; ROUKES, M. L.

ABSTRACT

NANOMECHANICAL RESONATORS CAN NOW BE REALIZED THAT ACHIEVE FUNDAMENTAL RESONANCE FREQUENCIES EXCEEDING 1 GHz, WITH QUALITY FACTORS (Q) IN THE RANGE $10^3 \leq Q \leq 10^5$. THE MINUSCULE ACTIVE MASSES OF THESE DEVICES, IN CONJUNCTION WITH THEIR HIGH QS, TRANSLATE INTO UNPRECEDENTED INERTIAL MASS SENSITIVITIES. THIS MAKES THEM NATURAL CANDIDATES FOR A VARIETY OF MASS SENSING APPLICATIONS. HERE WE EVALUATE THE ULTIMATE MASS SENSITIVITY LIMITS FOR NANOMECHANICAL RESONATORS OPERATING IN VACUO THAT ARE IMPOSED BY A NUMBER OF FUNDAMENTAL PHYSICAL NOISE PROCESSES. OUR ANALYSES INDICATE THAT NANOMECHANICAL RESONATORS OFFER IMMENSE POTENTIAL FOR MASS SENSING-ULTIMATELY WITH RESOLUTION AT THE LEVEL OF INDIVIDUAL MOLECULES.

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Our Changing Visual Environment

by Maureen MacKenzie-Taylor

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As technology changes, so too does the way we use information. Maureen MacKenzie-Taylor writes about these changes, and what they may mean for information design, especially when designing for younger generations.

In the TV, video and computer dominated world of the 1990s our visual culture is undergoing rapid change—a change which may well prove to be as monumental as was the change initiated by the invention of movable type back in the 15th century.

THE HISTORICAL CONTEXT

For over four hundred years the printed word dominated western communication, and western reading habits developed through continuous exposure to information presented as a block of texts. Western cultures came to accept the printed word as the hallmark of what distinguished them from pre-literate societies, and to see printed words as ‘containing’ the meaning. This led to a simplistic view of the communication process—printed documents were believed to be animate, able to ‘transmit’ their meanings to a reader who simply ‘received’ the meaning. Modernist thinking encouraged the belief that if the message was presented clearly and legibly then communication would take place. The work that was done in Europe by the Bauhaus and Swiss typography movement are excellent examples of this clean, clear approach to graphic communication.

Over the past decade, however, our understanding of the communication process has evolved beyond this simplistic sender/receiver ‘conduit’ view to an understanding of communication as a much more complex and uncertain process. Postmodern and constructionist theory allow us to understand that meaning can only be generated when the reader engages with the printed words. We now understand that meaning, and thus communication,

is brought into being by the interactive relationship between the reader and the document.

TECHNOLOGICAL CHANGE

Today’s documents look very different to the printed material of the last four centuries, and the printed word is no longer the dominant medium. Television, video and the rapidly evolving computerised multi-media publishing technologies are integral to our information-soaked world. In our contemporary visual language the influence of the computer is inescapable. The new technologies enable individuals to edit and compose texts, to liberate type from the block-like structures imposed by the rectangular metal printing frame, to create and layer type, to juxtapose images and sound, to produce and to publish. In effect, the new technologies give late 20th century desktop publishers and multi-media experts the same integrated control over visible language that was available to Renaissance man before the technology of moveable type created the separate and specialised professions of type cutter, author, artist, illustrator, designer, printer and publisher.

This integrated technological control over visual language permits an unprecedented freedom in giving a visual form to words. Typography remains a primary medium for communication, but type is no longer static and cast in metal. Type is malleable, constantly evolving—a product of rapidly evolving technology which in turn is a product of a rapidly changing society and culture.

The digital technology of recent decades has allowed a revolution in the look of the printed page. New generations of computer designers are challenging the traditional typographic assumptions of previous generations, by turning text into a set of visual elements to be taken apart, visually interfered with and reconstructed in a way that challenges readers to put it together for themselves to figure out the meanings. This ‘deconstructionist typography’ (see the example on the front page of this issue by Ray Gun typographer Davis Carson) creates a much more dynamic

printed page, with surprising juxtapositions and simultaneous layerings of disparate information. The proliferation of new magazines teach us to open anywhere, to read from back to front, to read just pictures and captions, or to read text and marginalia independently as well as together. Style magazines pioneering deconstructionist typography, such as *Émigré*, *The Face*, *Ray Gun* and *Bikini*, began on the margins, but now occupy the mainstream of youth culture. What would have appeared unapproachable, meaningless clutter a decade ago is now a preferred leisure reading material for a particular youth sub-culture. Could it be that as our visual environment is changing, so people are learning new reading habits?

Younger readers, accustomed to the saturated integration of text and image through the media of fast-paced television, film clips and computer and video games, are familiar with, and seem to cope with, a vast range of diverse visual material. From personal observations and informal research with educators in secondary and tertiary institutions it appears likely that the 'tv generation' are developing different ways of reading and processing visual information to preceding generations. Not restricted to reading sequentially and continuously, the Post-modern and post-Post-modern generations appear to have developed what we might call 'quick grab' literacy-the ability to absorb multiple hybrid visual codes and quick bytes of instant information. To date, little research has been published discussing how teenagers are interacting with deconstructed printed texts. We do not yet understand whether the younger generation is developing new ways of constructing meaning from such texts, or indeed, whether they even do generate meaning.

NEW TYPOGRAPHY AND LEGIBILITY

Classic studies in legibility research (Tinker, 1963, 1969; Spencer, 1968) provided fairly sound guidelines as to what combination of typographic variables would make visible language clear, easy to read, and least tiring to the eye. For the last thirty years information designers have followed these guidelines, carefully ma-

nipulating four typographic variables: the style of letterform, the size of letters, the length of line and the amount of space between lines, to ensure optimum legibility of texts, believing that a clear and legible text would ensure that communication takes place.

However, at the Institute we have long been aware that there are no universal fixed and finite 'cookbook' solutions to legibility and communication. Our research is showing that people read and understand best what they are most used to (MacKenzie, 1992:54-56).

In contrast, the earlier legibility studies suggested that legibility was somehow biologically determined. It was inferred that because of the way our eyes are constructed, and the ways in which people read with jumping eye movements, that serified text is easier to read because those little finishing strokes to letters aid the eye's progression from letter to letter, and improve reader's differentiation of word shapes.

In contrast to this earlier research, our recent testing of documents shows that in the 1990s there is no difference per se in ease and speed of reading and consequent understanding between serif and sanserif letterforms. What our results suggest is that legibility is culturally learned and sub-culturally developed. In the sixties readers were simply unfamiliar with the new modernist sanserif letterforms, and therefore found them harder to read. In the nineties, the general public are now as accustomed to sanserif letterforms as they are to serif letters. This is further confirmation of our hypothesis that people read best what they are most familiar with. As Zuzana Licko of *Émigré Graphics*, and creator of many deconstructed typeforms claims, 'no type face is inherently illegible; it is the reader's familiarity with faces that accounts for their legibility' (Smart, 1994).

Our visual environment is becoming increasingly complex. Computer printouts, OCR barcodes and digital neon signs are now familiar landmarks in our visual environment. (Remember how hard it was to read that first computer-generated bill?)

But while younger readers cope well with, and even seem to enjoy the interaction with new fragmented typography, older generations of readers, conditioned to the static simplicity of traditional printed pages of continuous text, are disinclined to make the necessary effort to reconstruct a text. In informal exploratory research, older readers dismisses the multiple, fragmented, disrupted texts of the new typography as 'illegible' and 'meaningless'.

From research that has been published on the processing of visual information (Marr, 1982) it would seem that 'quick grab' legibility demands a different mental activity to reading continuous text. This kind of activity—of synthesising a sequence of partial texts into a coherent whole—is in fact a fundamental cognitive skill—the on-going process through which we construct a coherent sense of our immediate environment through a combination of successive glances. As fragmented typography, with its bombardment of simultaneous layering and surprising juxtapositions, becomes more prevalent and inescapable, so it is highly likely that it will become more familiar to, and therefore more understandable to older readers. Every generation has the capacity to read both traditional continuous text and multi-layered dynamic texts, but each generation reads best what they are most familiar with.

APPROPRIATE TYPOGRAPHY

Optimum legibility is therefore conditional on the context of use and the reading habits of those who will use the information. What readers find acceptable in one genre of document, such as a newspaper, or insurance policy is not appropriate in either a novel or a youth culture magazine. If we want to make effective documents, we must use typography appropriately to meet the needs of our target audience and the context in which our documents will be used.

Knowing who you are designing for, knowing what their information needs are, determining how they need to access the information, how they need to act upon that information, and what they will find

visually inviting, are the critical factors in providing an appropriate visual solution.

The problem to be solved here is that there is little contemporary research on how visual understanding is evolving in response to our fast changing visual environment. What are the needs of readers in this fast-changing visual world? How effective is multi-media as a tool of communication? Is CD-ROM as promising an educational tool as the techno-enthusiasts proclaim? Do we need to rethink our models of information design to encourage interaction with our readers? Do we need, as Emery (1993) an eminent Australian typographer suggests, 'to provoke the reader into becoming an active participant in the construction of the message'?

In an earlier edition of this newsletter (MacKenzie, 1993) I argued that all printed information, no matter how mundane, should be both functional and appeal to us aesthetically. We can use typography to evoke the reader's response 'this is a desirable document which deserves to be read', rather than the reaction 'this is a lot of insignificant rubbish which I don't need to read'. By using the appearance of a document to engage the reader and sustain their interest, appearance can add to functionality.

However, because the appearance of visual information has changed so radically this past decade, traditional genres and ways of presenting particular types of information are most likely becoming inappropriate to a growing percentage of younger readers. There is an urgent need for research here, as this is the very real challenge for us as information designers in the 1990s. To ensure that the documents we produce are relevant to target audiences across the generations we need to understand the changing processes involved in visual understanding. For our visual language in information design to remain relevant in today's world we must change as our cultural world changes and reconcile conservative methods of typography, for so long integral to the world of information design, with the changes brought about by new cultural and technological stimuli.

I am not advocating attention grabbing techniques that force readers to struggle to create meaning, or collisions of type and images, designed to jar reader perceptions and in this way to compel attention. Nor am I arguing, as Gary Emery has argued (1993:10-11) that typographers today must 'confront television and film media on their own terms and compete aggressively for attention'. Rather, I want to stress the importance of understanding the effect that rapidly changing technology and its effect on our visual environment has on the way different generations process visual information.

To be effective communicators, our visual language must reflect the hybrid heterogeneous modern world of the 1990s, but to be able to contribute effectively to communication in the digital age, we need to better understand the processes that enable legibility and the generation of meaning. Most particularly, we need to understand how the new generations of consumers and citizens interact with a text. Are they developing new ways of reading? Are they developing new ways of creating meaning? Why and how is fragmented deconstructed type and text effective for this generation? If we make use of the quick grab legibility does that exclude older readers from understanding?

For millions of years, mankind lived just like the animals. Then something happened which unleashed the power of our imagination. We learned to talk and we learned to listen...

Stephen William Hawking (born 8 January 1942) British physicist

Where there is life, there is hope. HKUST conference 15 June 2006

Many people would claim that the boundary conditions are not part of physics but belong to metaphysics or religion. They would claim that nature had complete freedom to start the universe off any way it wanted. That may be so, but it could also have made it evolve in a completely arbitrary and random manner. Yet all the evidence is that it evolves in a regular way according to certain laws. It would therefore seem reasonable to suppose that there are also laws governing the boundary conditions.

"The Quantum State of the Universe", Nuclear Physics (1984)

There ought to be something very special about the boundary conditions of the universe and what can be more special than that there is no boundary?

As quoted in The Anthropic Cosmological Principle (1986) by John D. Barrow and Frank J. Tipler.

What I have done is to show that it is possible for the way the universe began to be determined by the laws of science. In that case, it would not be necessary to appeal to God to decide how the universe began. This doesn't prove that there is no God, only that God is not necessary.

Der Spiegel (17 October 1988)

We are just an advanced breed of monkeys on a minor planet of a very average star. But we can understand the Universe. That makes us something very special.

Der Spiegel (17 October 1988)

For millions of years, mankind lived just like the animals. Then something happened which unleashed the power of our imagination. We learned to talk and we learned to listen. Speech has allowed the communication of ideas, enabling human beings to work together to build the impossible. Mankind's greatest achievements have come about by talking, and its

greatest failures by not talking. It doesn't have to be like this. Our greatest hopes could become reality in the future. With the technology at our disposal, the possibilities are unbounded. All we need to do is make sure we keep talking.

British Telecom advertisement (1993), part of which was used in Pink Floyd's *The Division Bell* (1994).

I don't believe that the ultimate theory will come by steady work along existing lines. We need something new. We can't predict what that will be or when we will find it because if we knew that, we would have found it already! It could come in the next 20 years, but we might never find it.

Science Watch (September 1994)

All of my life, I have been fascinated by the big questions that face us, and have tried to find scientific answers to them. Perhaps that is why I have sold more books on physics than Madonna has on sex.

The Illustrated A Brief History of Time (1996)

Although September 11 was horrible, it didn't threaten the survival of the human race, like nuclear weapons do.

Interview "Colonies in space may be only hope, says Hawking" by Roger Highfield in *Daily Telegraph* (16 October 2001)

I don't think the human race will survive the next thousand years, unless we spread into space. There are too many accidents that can befall life on a single planet. But I'm an optimist. We will reach out to the stars.

"Colonies in space may be only hope, says Hawking" by Roger Highfield in *Daily Telegraph* (16 October 2001)

I'm sorry to disappoint science fiction fans, but if information is preserved, there is no possibility of using black holes to travel to other universes. If you jump into a black hole, your mass energy will be returned to our universe but in a mangled form which contains the information about what you were like but in a state where it can not be easily recognized. It is like burning an encyclopedia. Information is not lost, if one keeps the smoke and the ashes. But it is difficult to read. In practice, it would be too difficult to re-build a macroscopic object like an encyclopedia that fell inside a black hole from information in the radiation, but the

information preserving result is important for microscopic processes involving virtual black holes.

"Information Loss in Black Holes" (July 2005)

So Einstein was wrong when he said "God does not play dice". Consideration of black holes suggests, not only that God does play dice, but that He sometimes confuses us by throwing them where they can't be seen.

"The Nature of Space and Time" (online text)

Unsourced variants: Not only does God play dice with the Universe; he sometimes casts them where they can't be seen. _ Not only does God play dice, but... he sometimes throws them where they cannot be seen.

I have no idea. People who boast about their IQ are losers.

Response upon being questioned as to his IQ, in interview with Deborah Solomon "The Science of Second-Guessing", *The New York Times* (12 December 2004)

Life would be tragic if it weren't funny.

Interview with Deborah Solomon "The Science of Second-Guessing", *The New York Times* (12 December 2004)

My expectations were reduced to zero when I was 21. Everything since then has been a bonus.

Interview with Deborah Solomon "The Science of Second-Guessing", *The New York Times* (12 December 2004) Unsourced variant: "When one's expectations are reduced to zero, one really appreciates everything one does have."

Evolution has ensured that our brains just aren't equipped to visualise 11 dimensions directly. However, from a purely mathematical point of view it's just as easy to think in 11 dimensions, as it is to think in three or four.

"Return of the time lord" interview in *The Guardian* (27 September 2005)

I think that it's important for scientists to explain their work, particularly in cosmology. This now answers many questions once asked of religion.

"Return of the time lord" interview in *The Guardian* (27 September 2005)

It is a waste of time to be angry about my disability. One has to get on with life and I haven't done badly. People won't have time for you if you are always angry or complaining.

"Return of the time lord" interview in The Guardian (27 September 2005)

A Brief History of Time (1988)

Someone told me that each equation I included in the book would halve the sales. I therefore resolved not to have any equations at all. In the end, however, I did put in one equation, Einstein's famous equation, $E = mc^2$. I hope that this will not scare off half of my potential readers.

Any physical theory is always provisional, in the sense that it is only a hypothesis: you can never prove it. No matter how many times the results of experiments agree with some theory, you can never be sure that the next time the result will not contradict the theory. On the other hand, you can disprove a theory by finding even a single observation that disagrees with the predictions of the theory. As philosopher of science Karl Popper has emphasized, a good theory is characterized by the fact that it makes a number of predictions that could in principle be disproved or falsified by observation. Each time new experiments are observed to agree with the predictions the theory survives, and our confidence in it is increased; but if ever a new observation is found to disagree, we have to abandon or modify the theory.

It has certainly been true in the past that what we call intelligence and scientific discovery have conveyed a survival advantage. It is not so clear that this is still the case: our scientific discoveries may well destroy us all, and even if they don't, a complete unified theory may not make much difference to our chances of survival. However, provided the universe has evolved in a regular way, we might expect that the reasoning abilities that natural selection has given us would be valid also in our search for a complete unified theory, and so would not lead us to the wrong conclusions.

Bodies like the earth are not made to move on curved orbits by a force called gravity; instead, they follow the nearest thing to a straight path in a curved space, which is called a geodesic. A geodesic is the shortest (or longest) path between two nearby points.

The whole history of science has been the gradual realization that events do not happen

in an arbitrary manner, but that they reflect a certain underlying order, which may or may not be divinely inspired.

One could say: "The boundary condition of the universe is that it has no boundary." The universe would be completely self-contained and not affected by anything outside itself. It would neither be created nor destroyed. It would just BE.

Just like a computer, we must remember things in the order in which entropy increases. This makes the second law of thermodynamics almost trivial. Disorder increases with time because we measure time in the direction in which disorder increases. You can't have a safer bet than that!

Maybe that is our mistake: maybe there are no particle positions and velocities, but only waves. It is just that we try to fit the waves to our preconceived ideas of positions and velocities. The resulting mismatch is the cause of the apparent unpredictability.

Even if there is only one possible unified theory, it is just a set of rules and equations. What is it that breathes fire into the equations and makes a universe for them to describe? The usual approach of science of constructing a mathematical model cannot answer the questions of why there should be a universe for the model to describe. Why does the universe go to all the bother of existing?

If we do discover a complete theory, it should in time be understandable in broad principle by everyone, not just a few scientists. Then we shall all, philosophers, scientists, and just ordinary people, be able to take part in the discussion of the question of why it is that we and the universe exist. If we find the answer to that, it would be the ultimate triumph of human reason – for then we would know the mind of God.

Last lines. Hawking later wrote: "In the proof stage I nearly cut the last sentence in the book... Had I done so, the sales might have been halved. Black Holes and Baby Universes and Other Essays (1993)

The Beginning of Time

When I gave a lecture in Japan, I was asked not to mention the possible re-collapse of the universe, because it might affect the stock market. However, I can re-assure anyone who is nervous about their investments that it is a bit early to sell: even if the universe does come

to an end, it won't be for at least twenty billion years. By that time, maybe the GATT trade agreement will have come into effect.

The Steady State theory was what Karl Popper would call a good scientific theory: it made definite predictions, which could be tested by observation, and possibly falsified. Unfortunately for the theory, they were falsified.

To show this diagram properly, I would really need a four dimensional screen. However, because of government cuts, we could manage to provide only a two dimensional screen.

The universe would have expanded in a smooth way from a single point. As it expanded, it would have borrowed energy from the gravitational field, to create matter. As any economist could have predicted, the result of all that borrowing, was inflation. The universe expanded and borrowed at an ever-increasing rate. Fortunately, the debt of gravitational energy will not have to be repaid until the end of the universe.

Attributed

Equations are just the boring part of mathematics. I attempt to see things in terms of geometry.

He's a Platonist and I am a positivist. He's worried that Schrödinger's cat is in a quantum state, where it is half dead and half alive. He feels that can't correspond to reality. But that doesn't bother me. I don't demand that a theory correspond to reality because I do not know what it is. Reality is not a quality you can test with litmus paper. All I am concerned with is that the theory should predict the results of measurements. Quantum theory does this very successfully. It predicts that the result of an observation is either that the cat is alive or that it is dead. It is like you can't be slightly pregnant; you either are or you aren't.

About Roger Penrose

I find that American & Scandinavian accents work better with women.

Response to a question about the American accent of his synthesiser

I have hardly ever known a mathematician who was able to reason.

I have noticed even people who claim everything is predestined, and that we can do nothing to change it, look before they cross the

road.

I think computer viruses should count as life. I think it says something about human nature that the only form of life we have created so far is purely destructive. We've created life in our own image.

I'm working on that.

Smilingly looking at the Warp Core engine of the starship Enterprise on the set of Star Trek: The Next Generation, in the episode in which he plays himself.

It is not clear that intelligence has any long-term survival value.

It matters if you just don't give up.

Maybe my variety is due to bad absorption of vitamins.

After being asked by the British Medical Journal why his condition had evolved differently from a typical case of motor neurone disease.

My goal is simple. It is a complete understanding of the universe, why it is as it is and why it exists at all.

The greatest enemy of knowledge is not ignorance, it is the illusion of knowledge.

The world has changed far more in the past 100 years than in any other century in history. The reason is not political or economic but technological – technologies that flowed directly from advances in basic science.

There are grounds for cautious optimism that we may now be near the end of the search for the ultimate laws of nature.

To confine our attention to terrestrial matters would be to limit the human spirit.

We could call order by the name of God, but it would be an impersonal God. There's not much personal about the laws of physics.

We shouldn't be surprised that conditions in the universe are suitable for life, but this is not evidence that the universe was designed to allow for life.

When I hear of Schrödinger's cat, I reach for my gun.

A variation on a line from German playwright

Hanns Johst's play *Schlageter*: "Wenn ich Kultur höre ... entsichere ich meinen Browning" - "Whenever I hear of culture... I release the safety-catch of my Browning!" (Act 1, Scene 1).

It is clear to everyone that astronomy at all events compels the soul to look upwards, and draws it from the things of this world to the other. - Plato

Space isn't remote at all. It's only an hour's drive away if your car could go straight upwards. - Fred Hoyle

Interestingly, according to modern astronomers, space is finite. This is a very comforting thought— particularly for people who can never remember where they have left things. - Woody Allen

Space is big. You just won't believe how vastly, hugely, mind- bogglingly big it is. I mean, you may think it's a long way down the road to the drug store, but that's just peanuts to space. - Douglas Adams

There is just one thing I can promise you about the outer-space program: your dollar will go further. - Wernher Von Braun

I'm from Iowa, I only work in space. - James T. Kirk

Astronomy is perhaps the science whose discoveries owe least to chance, in which human understanding appears in its whole magnitude, and through which man can best learn how small he is. - G. C. Lichtenberg

The poet alone knows astronomy, chemistry, vegetation, and animation, for he does not stop at these facts, but employs them as signs. He knows why the plain, or meadow of space, was strewn with these flowers we call suns, and moons, and stars; why the great deep is adorned with animals, with men, and gods; for, in every word he speaks he rides on them as the horses of thought. - Ralph Waldo Emerson

Only two things are infinite, the universe and human stupidity, and I'm not sure about the former. - Albert Einstein

There is a theory which states that if ever anybody discovers exactly what the Universe is for and why it is here, it will instantly disappear and be replaced by something even more bizarre and inexplicable. There is another theory which states that this has already happened. - Douglas Adams

In answer to the question of why it happened, I offer the modest proposal that our Universe is simply one of those things which happen from time to time. - Edward P. Tryon

The universe is a big place, perhaps the biggest. - Kilgore Trout, (Philip Jose Farmer)

I'm astounded by people who want to 'know' the universe when it's hard enough to find your way around Chinatown. - Woody Allen

In the beginning the Universe was created. This has made a lot of people very angry and been widely regarded as a bad move. - Douglas Adams

My sense of God is my sense of wonder about the Universe. - Albert Einstein

We shouldn't be surprised that conditions in the universe are suitable for life, but this is not evidence that the universe was designed to allow for life. - Stephen Hawking

The universe seems neither benign nor hostile, merely indifferent. - Carl Sagan

Deep in the human unconscious is a pervasive need for a logical universe that makes sense. But the real universe is always one step beyond logic. - Frank Herbert

Computer Science is no more about computers than astronomy is about telescopes. - E. W. Dijkstra

We are all in the gutter, but some of us are looking at the stars. - Oscar Wilde

Ideals are like stars: you will not succeed in touching them with your hands, but like the seafaring man on the ocean desert of waters, you choose them as your guides, and following them, you reach your destiny. - Carl Schurz

When you reach for the stars you may not quite get one, but you won't come up with a handful of mud either. - Leo Burnett

Two men look out through the same bars: One sees the mud and one the stars. - Frederick Langbridge

I ask you to look both ways. For the road to a knowledge of the stars leads through the atom; and important knowledge of the atom has been reached through the stars. - Sir Arthur Eddington

If we long to believe that the stars rise and set for us, that we are the reason there is a Universe, does science do us a disservice in deflating our conceits? - Carl Sagan

The way to the stars is open. - Sergei Koroljov

Be humble, for the worst thing in the world is of the same stuff as you; be confident, for the stars are of the same stuff as you. - Nicholai Velimirovic

Keep your eyes on the stars, and your feet on the ground. - Theodore Roosevelt

When it is dark enough, you can see the stars. - Ralph Waldo Emerson

Shoot for the moon. Even if you miss it you will land among the stars. - Les Brown

Be glad of life because it gives you the chance to love, and to work, and to play and to look up at the stars. - Henry Van Dyke

For my part I know nothing with any certainty, but the sight of the stars makes me dream. - Vincent Van Gogh

When we are chafed and fretted by small cares, a look at the stars will show us the littleness of our own interests. - Mara Mitchell

Too low they build, who build beneath the stars. - Edward Young

In the vast reaches of the dry, cold night, thousands of stars were constantly appearing, and their sparkling icicles, loosened at once, began to slip gradually toward the horizon. - Albert Camus

The cosmos is a gigantic flywheel making 10,000 revolutions per minute. Man is a sick fly taking a dizzy ride on it. - H. L. Mencken

Ignorance is the night of the mind, but a night without moon and star. - Confucius

We choose to go to the Moon in this decade and do the other things, not because they are easy - but because they are hard! Because that challenge is one we are willing to accept, one we are unwilling to postpone and one we intend to win! - John F. Kennedy

Programming today is a race between software engineers striving to build bigger and better idiot-proof programs, and the Universe trying to produce bigger and better idiots. So far, the

Universe is winning. - Rich Cook

There is a coherent plan in the universe, though I don't know what it's a plan for. - Fred Hoyle

Sometimes I think the surest sign that intelligent life exists elsewhere in the universe is that none of it has tried to contact us. - Bill Watter-son

We are an impossibility in an impossible universe. - Ray Bradbury

My theology, briefly, is that the universe was dictated but not signed. - Christopher Morley

I'm worried that the universe will soon need replacing. It's not holding a charge. - Edward Chilton

Everything you've learned in school as "obvious" becomes less and less obvious as you begin to study the universe. For example, there are no solids in the universe. There's not even a suggestion of a solid. There are no absolute continuums. There are no surfaces. There are no straight lines. - R. Buckminster Fuller

The effort to understand the universe is one of the very few things that lifts human life a little above the level of farce, and gives it some of the grace of tragedy. - Steven Weinberg

When they discover the center of the universe, a lot of people will be disappointed to discover they are not it. - Bernard Bailey

If you want to make an apple pie from scratch, you must first create the universe. - Carl Sagan

It is my supposition that the Universe in not only queerer than we imagine, is queerer than we CAN imagine. - J.B.S. Haldane

Humor is just another defense against the universe. - Mel Brooks

Wonder is what sets us apart from other life forms. No other species wonders about the meaning of existence or the complexity of the universe or themselves. - Herbert W. Boyer

We are just an advanced breed of monkeys on a minor planet of a very average star. But we can understand the Universe. That makes us something very special. - Stephen Hawking

The universe is full of magical things, patiently waiting for our wits to grow sharper. - Eden Phillpotts

A penny will hide the biggest star in the Universe if you hold it close enough to your eye.
- Samuel Grafton

We know next to nothing about virtually everything. It is not necessary to know the origin of the universe; it is necessary to want to know. Civilization depends not on any particular knowledge, but on the disposition to crave knowledge. - George F. Will

Not only does God play dice with the Universe - he sometimes casts them where they can't be seen. - Stephen Hawking

My goal is simple. It is a complete understanding of the universe, why it is as it is and why it exists at all. - Stephen Hawking

"I cannot conceive of a personal God who would directly influence the actions of individuals, or would directly sit in judgment on creatures of his own creation. I cannot do this in spite of the fact that mechanistic causality has, to a certain extent, been placed in doubt by modern science. [He was speaking of Quantum Mechanics and the breaking down of determinism.] My religiosity consists in a humble admiration of the infinitely superior spirit that reveals itself in the little that we, with our weak and transitory understanding, can comprehend of reality. Morality is of the highest importance – but for us, not for God."

[Albert Einstein, from "Albert Einstein: The Human Side", edited

by Helen Dukas and Banesh Hoffman, Princeton University Press]

"If people are good only because they fear punishment, and hope for reward, then we are a sorry lot indeed."

[Albert Einstein]

"I cannot conceive of a God who rewards and punishes his creatures, or has a will of the kind that we experience in ourselves. Neither can I nor would I want to conceive of an individual that survives his physical death; let feeble souls, from fear or absurd egoism, cherish such thoughts. I am satisfied with the mystery of the eternity of life and with the awareness and a glimpse of the marvelous structure of the existing world, together with the devoted striving to comprehend a portion, be it ever so tiny, of the Reason that manifests itself in nature."

[Albert Einstein, _The World as I See It _]

"The idea of a personal God is an anthropological concept which I am unable to take seriously."

[Albert Einstein, letter to Hoffman and Dukas, 1946]

"The foundation of morality should not be made dependent on myth nor tied to any authority lest doubt about the myth or about the legitimacy of the authority imperil the foundation of sound judgment and action."

[Albert Einstein]

"I do not believe in immortality of the individual, and I consider ethics to be an exclusively human concern with no superhuman authority behind it."

["Albert Einstein: The Human Side", edited by Helen Dukas and Banesh

Hoffman, and published by Princeton University Press.]

"I have repeatedly said that in my opinion the idea of a personal God is a childlike one, but I do not share the crusading spirit of the professional atheist whose fervor is mostly due to a painful act of liberation from the fetters of religious indoctrination received in youth. I prefer an attitude of humility corresponding to the weakness of our intellectual understanding of nature and of our own being."

[Albert Einstein]

"A man's ethical behavior should be based effectually on sympathy, education, and social ties and needs; no religious basis is necessary. Man would indeed be in a poor way if he had to be restrained by fear of punishment and hope of reward after death."

[Albert Einstein, "Religion and Science", New York Times Magazine, 9 November 1930]

"What I see in Nature is a magnificent structure that we can comprehend only very imperfectly, and that must fill a thinking person with a feeling of "humility." This is a genuinely religious feeling that has nothing to do with mysticism"

[Albert Einstein]

"I cannot imagine a God who rewards and punishes the objects of his creation, whose purposes are modeled after our own – a God, in short,

who is but a reflection of human frailty. Neither can I believe that the individual survives the death of his body, although feeble souls harbor such thoughts through fear or ridiculous egotisms.”

[Albert Einstein, obituary in New York Times, 19 April 1955]

“Although I cannot believe that the individual survives the death of his body, feeble souls harbor such thought through fear or ridiculous egotism.”

[Albert Einstein]

“It was, of course, a lie what you read about my religious convictions, a lie which is being systematically repeated. I do not believe in a personal God and I have never denied this but have expressed it clearly. If something is in me which can be called religious then it is the unbounded admiration for the structure of the world so far as our science can reveal it.”

[Albert Einstein, 1954, from “Albert Einstein: The Human Side”, edited by Helen Dukas and Banesh Hoffman, Princeton University Press]

“The mystical trend of our time, which shows itself particularly in the rampant growth of the so-called Theosophy and Spiritualism, is for me no more than a symptom of weakness and confusion. Since our inner experiences consist of reproductions, and combinations of sensory impressions, the concept of a soul without a body seem to me to be empty and devoid of meaning.”

[Albert Einstein]

“I worry that, especially as the Millennium edges nearer, pseudoscience and superstition will seem year by year more tempting, the siren song of unreason more sonorous and attractive. Where have we heard it before? Whenever our ethnic or national prejudices are aroused, in times of scarcity, during challenges to national self-esteem or nerve, when we agonize about our diminished cosmic place and purpose, or when fanaticism is bubbling up around us—then, habits of thought familiar from ages past reach for the controls.

“The candle flame gutters. Its little pool of light trembles. Darkness gathers. The demons begin to stir.”

[Carl Sagan, “The Demon-Haunted World: Science As a Candle in the Dark”]

“Life is but a momentary glimpse of the wonder of this astonishing universe, and it is sad to see so many dreaming it away on spiritual fantasy.”

[Carl Sagan]

“(When asked merely if they accept evolution, 45 percent of Americans say yes. The figure is 70 percent in China.) When the movie “Jurassic Park” was shown in Israel, it was condemned by some Orthodox rabbis because it accepted evolution and because it taught that dinosaurs lived a hundred million years ago—when, as is plainly stated at every Rosh Hashonhan and every Jewish wedding ceremony, the Universe is less than 6,000 years old.”

[Carl Sagan, _The Demon-Haunted World:

Science as a Candle in the Dark_, p. 325]

“In science it often happens that scientists say, ‘You know that’s a really good argument; my position is mistaken,’ and then they actually change their minds and you never hear that old view from them again. They really do it. It doesn’t happen as often as it should, because scientists are human and change is sometimes painful. But it happens every day. I cannot recall the last time something like that happened in politics or religion.”

[Carl Sagan, 1987 CSICOP keynote address]

“One of the saddest lessons of history is this: If we’ve been bamboozled long enough, we tend to reject any evidence of the bamboozle. We’re no longer interested in finding out the truth. The bamboozle has captured us. It is simply too painful to acknowledge – even to ourselves – that we’ve been so credulous. (So the old bamboozles tend to persist as the new bamboozles rise.)”

[Carl Sagan, “The Fine Art of Baloney Detection,”]

“At the extremes it is difficult to distinguish pseudoscience from rigid, doctrinaire religion.”

[Carl Sagan, “The Demon-Haunted World: Science As a Candle in the Dark”]

“If you want to save your child from polio, you can pray or you can inoculate....Try science.”

[Carl Sagan, quoted in “2000 Years of Disbelief, Famous People with the Courage to Doubt”, by James A. Haught, Prometheus Books, 1996]

"The idea that God is an oversized white male with a flowing beard who sits in the sky and tallies the fall of every sparrow is ludicrous. But if by "God" one means the set of physical laws that govern the universe, then clearly there is such a God. This God is emotionally unsatisfying... it does not make much sense to pray to the law of gravity."

[Carl Sagan]

"Is it fair to be suspicious of an entire profession because of a few bad apples? There are at least two important differences, it seems to me. First, no one doubts that science actually works, whatever mistaken and fraudulent claim may from time to time be offered. But whether there are *any* miraculous cures from faith-healing, beyond the body's own ability to cure itself, is very much at issue. Secondly, the expose' of fraud and error in science is made almost exclusively by science. But the exposure of fraud and error in faith-healing is almost never done by other faith-healers."

[Carl Sagan, "The Demon-Haunted World: Science As a Candle in the Dark"]

"Many statements about God are confidently made by theologians on grounds that today at least sound specious. Thomas Aquinas claimed to prove that God cannot make another God, or commit suicide, or make a man without a soul, or even make a triangle whose interior angles do not equal 180 degrees. But Bolyai and Lobachevsky were able to accomplish this last feat (on a curved surface) in the nineteenth century, and they were not even approximately gods."

[Carl Sagan, _Broca's Brain_]

"Think of how many religions attempt to validate themselves with prophecy. Think of how many people rely on these prophecies, however vague, however unfulfilled, to support or prop up their beliefs. Yet has there ever been a religion with the prophetic accuracy and reliability of science?"

[Carl Sagan, "The Demon-Haunted World: Science As a Candle in the Dark"]

"I would love to believe that when I die I will live again, that some thinking, feeling, remembering part of me will continue. But as much as I want to believe that, and despite the ancient and worldwide cultural traditions that assert an afterlife, I know of nothing to suggest that it is more than wishful thinking."

[Carl Sagan]

"Skeptical scrutiny is the means, in both science and religion, by which deep thoughts can be winnowed from deep nonsense."

[Carl Sagan]

"You can't convince a believer of anything; for their belief is not based on evidence, it's based on a deep seated need to believe."

[Carl Sagan]

"If we long to believe that the stars rise and set for us, that we are the reason there is a Universe, does science do us a disservice in deflating our conceits?....For me, it is far better to grasp the Universe as it really is than to persist in delusion, however satisfying and reassuring."

[Carl Sagan, "The Demon-Haunted World: Science As a Candle in the Dark"]

"Finding the occasional straw of truth awash in a great ocean of confusion and bamboozle requires intelligence, vigilance, dedication and courage. But if we don't practice these tough habits of thought, we cannot hope to solve the truly serious problems that face us – and we risk becoming a nation of suckers, up for grabs by the next charlatan who comes along."

[Carl Sagan, "The Fine Art of Baloney Detection,"]

"I maintain there is much more wonder in science than in pseudoscience. And in addition, to whatever measure this term has any meaning, science has the additional virtue, and it is not an inconsiderable one, of being true."

[Carl Sagan, The Burden Of Skepticism]

"You see, the religious people – most of them – really think this planet is an experiment. That's what their beliefs come down to. Some god or other is always fixing and poking, messing around with tradesmen's wives, giving tablets on mountains, commanding you to mutilate your children, telling people what words they can say and what words they can't say, making people feel guilty about enjoying themselves, and like that. Why can't the gods let well enough alone? All this intervention speaks of incompetence. If God didn't want Lot's wife to look back, why didn't he make her obedient, so she'd do what her husband told her? Or if he hadn't made Lot such a shithhead, maybe she

would have listened to him more. If God is omnipotent and omniscient, why didn't he start the universe out in the first place so it would come out the way he wants? Why's he constantly repairing and complaining? No, there's one thing the Bible makes clear: The biblical God is a sloppy manufacturer. He's not good at design, he's not good at execution. He'd be out of business if there was any competition."

[Carl Sagan, character Sol Hadden in *_Contact_*, 1985]

"There was no deathbed conversion," Druyan says. "No appeals to God, no hope for an after-life, no pretending that he and I, who had been inseparably for twenty years, were not saying goodbye forever."

"Didn't he want to believe?" she was asked.

"Carl never wanted to believe," she replies fiercely. "He wanted to KNOW."

[Ann Druyan, Carl Sagan's wife, from *News-week* magazine]

HAWKING

"What I have done is to show that it is possible for the way the universe began to be determined by the laws of science. In that case, it would not be necessary to appeal to God to decide how the universe began. This doesn't prove that there is no God, only that God is not necessary."

[Stephen W. Hawking, *Der Spiegel*, 1989]

DAWKINS

"It's been suggested that if the supernaturalists really had the powers they claim, they'd win the lottery every week. I prefer to point out that they could also win a Nobel Prize for discovering fundamental physical forces hitherto unknown to science. Either way, why are they wasting their talents doing party turns on television?"

[Richard Dawkins, *The Richard Dimbleby Lecture: Science, Delusion and the Appetite for Wonder*]

"Certainly I see the scientific view of the world as incompatible with religion, but that is not what is interesting about it. It is also incompatible with magic, but that also is not worth stressing. What is interesting about the scientific world view is that it is true, inspiring,

remarkable and that it unites a whole lot of phenomena under a single heading."

[Richard Dawkins]

"It is often said, mainly by the 'no-contests', that although there is no positive evidence for the existence of God, nor is there evidence against his existence. So it is best to keep an open mind and be agnostic. At first sight that seems an unassailable position, at least in the weak sense of Pascal's wager. But on second thoughts it seems a cop-out, because the same could be said of Father Christmas and tooth fairies. There may be fairies at the bottom of the garden. There is no evidence for it, but you can't *prove* that there aren't any, so shouldn't we be agnostic with respect to fairies?"

[Richard Dawkins]

"In childhood our credulity serves us well. It helps us to pack, with extraordinary rapidity, our skulls full of the wisdom of our parents and our ancestors. But if we don't grow out of it in the fullness of time, our ... nature makes us a sitting target for astrologers, mediums, gurus, evangelists, and quacks. We need to replace the automatic credulity of childhood with the constructive skepticism of adult science."

[Richard Dawkins]

"They express a preference for 'natural' methods of population limitation, and a natural method is exactly what they are going to get. It is called starvation."

[Richard Dawkins, "The Selfish Gene"]

"Science offers us an explanation of how complexity (the difficult) arose out of simplicity (the easy). The hypothesis of God offers no worthwhile explanation for anything, for it simply postulates what we are trying to explain. It postulates the difficult to explain, and leaves it at that. We cannot prove that there is no God, but we can safely conclude the He is very, very improbable indeed."

[Richard Dawkins, from the *_New Humanist_*, the Journal

of the Rationalist Press Association, Vol 107 No 2]

"The trouble is that God in this sophisticated, physicist's sense bears no resemblance to the God of the Bible or any other religion. If a physicist says God is another name for Planck's con-

stant, or God is a superstring, we should take it as a picturesque metaphorical way of saying that the nature of superstrings or the value of Planck's constant is a profound mystery. It has obviously not the smallest connection with a being capable of forgiving sins, a being who might listen to prayers, who cares about whether or not the Sabbath begins at 5pm or 6pm, whether you wear a veil or have a bit of arm showing; and no connection whatever with a being capable of imposing a death penalty on His son to expiate the sins of the world before and after he was born. "

[Richard Dawkins]

"If you have a faith, it is statistically overwhelmingly likely that it is the same faith as your parents and grandparents had. No doubt soaring cathedrals, stirring music, moving stories and parables, help a bit. But by far the most important variable determining your religion is the accident of birth. The convictions that you so passionately believe would have been a completely different, and largely contradictory, set of convictions, if only you had happened to be born in a different place. Epidemiology, not evidence."

[Richard Dawkins]

"Blind faith can justify anything. In a man believes in a different god, or even if he uses a different ritual for worshipping the same god, blind faith can decree that he should die - on the cross, at the stake, skewered on a Crusader's sword, shot in a Beirut street, or blown up in a bar in Belfast. Memes for blind faith have their own ruthless ways of propagating themselves. This is true of patriotic and political as well as religious blind faith."

[Richard Dawkins, "The Selfish Gene"]

"Faith is the great cop-out, the great excuse to evade the need to think and evaluate evidence. Faith is belief in spite of, even perhaps because of, the lack of evidence."

[Richard Dawkins]

"I suspect that today if you asked people to justify their belief in God, the dominant reason would be scientific. Most people, I believe, think that you need a God to explain the existence of the world, and especially the existence of life. They are wrong, but our education system is such that many people don't know it. "

[Richard Dawkins]

"The universe we observe has precisely the properties we should expect if there is, at bottom, no design, no purpose, no evil and no good, nothing but blind pitiless indifference."

[Richard Dawkins, "River Out of Eden"]

"On the contrary, if the universe were just electrons and selfish genes, meaningless tragedies like the crashing of this bus [full of children from a Roman Catholic school and for no apparent reason but with wholesale loss of life] are exactly what we should expect, along with equally meaningless *good* [italics in original] fortune. Such a universe would be neither evil nor good in intention. It would manifest no intentions of any kind. In a universe of blind physical forces and genetic replication, some people are going to get hurt, other people are going to get lucky, and you won't find any rhyme or reason in it, nor any justice. The universe we observe has precisely the properties we should expect if there is, at bottom, no design, no purpose, no evil, and no good, nothing but blind, pitiless indifference."

[Richard Dawkins, *River Out of Eden: A Darwinian View of*

Life, 1995, BasicBooks, New York; ISBN 0-465-01606-5]

"I suspect the reason is that most people [...] have a residue of feeling that Darwinian evolution isn't quite big enough to explain everything about life. All I can say as a biologist is that the feeling disappears progressively the more you read about and study what is known about life and evolution. I want to add one thing more. The more you understand the significance of evolution, the more you are pushed away from the agnostic position and towards atheism. Complex, statistically improbable things are by their nature more difficult to explain than simple, statistically probable things."

[Richard Dawkins, from the *New Humanist*, the Journal

of the Rationalist Press Association, Vol 107 No 2]

"The theory of evolution by cumulative natural selection is the only theory we know of that is in principle capable of explaining the existence of organized complexity."

[Richard Dawkins, *The Blind Watchmaker* (New York: W. W. Norton & Company, 1987), p. 317]

"And it's not just faith itself: it's the idea that faith is a virtue and the less evidence there is, the more virtuous it is. You can actually quote, well, Tertullian for example: "It is certain because it is impossible." Sir Thomas Brown, actually seeking for more difficult things to believe, because things for which there is mere evidence are just too easy, and it's no test of his faith. In order to have a test of your faith, you must be asked to believe really daft things like the transubstantiation, you know, the blood of Christ turning into wine, and stuff... That is so manifestly absurd that you've got to be a really great believer, in the class of the Electric Monk, in order to believe it..... You're actually showing off your believing credentials by the ability to believe something like that... If it were an easy thing to believe, substantiated by facts, then it wouldn't be any great achievement."

[Richard Dawkins, interview with Douglas Adams]

FREUD

"Religion is comparable to a childhood neurosis."

[Sigmund Freud, "Future of an Illusion"]

"In the long run, nothing can withstand reason and experience, and the contradiction religion offers to both is palpable."

[Sigmund Freud, Austrian physician and pioneer psychoanalyst (1856-1939)]

"While the different religions wrangle with one another as to which of them is in possession of the truth, In our view the truth of religion may be altogether disregarded...if one attempts to assign religion it's place in mans evolution, it seems not so much to be a lasting acquisition, as a parallel to the neurosis which the civilized individual must pass through on his way from childhood to maturity."

[Sigmund Freud, "Moses and Monotheism"]

"No, our science is no illusion. But an illusion it would be to suppose that what science cannot give us we can get elsewhere."

[Sigmund Freud, "The Future of an Illusion"]

"A great deal is already gained with the first step: the humanization of nature. Impersonal forces and destinies cannot be approached... if everywhere in nature there are Beings around

us of a kind that we know in our own society.... we can apply the same methods against these violent supermen outside that we employ in our own society; we can try to adjure them, to appease them, to bribe them, and, by so influencing them, we may rob them of a part of their power

[Sigmund Freud, "The Future of an Illusion"]

"When a man has once brought himself to accept uncritically all the absurdities that religious doctrines put before him and even to overlook the contradictions between them, we need not be greatly suprised at the weakness of his intellect"

[Sigmund Freud: The Future of an Illusion]

"Neither in my private life nor in my writings, have I ever made a secret of being an out-and-out unbeliever."

[Sigmund Freud, letter to Charles Singer]

"Demons do not exist any more than gods do, being only the products of the psychic activity of man."

[Sigmund Freud, New York Times Magazine, 6 May 1956]

"The more the fruits of knowledge become accessible to men, the more widespread is the decline of religious belief."

[Sigmund Freud]

OTHER SCIENTISTS

"Today, the theory of evolution is an accepted fact for everyone but a fundamentalist minority, whose objections are based not on reasoning but on doctrinaire adherence to religious principles."

[James Watson, winner of the Nobel prize for his co-discovery of the structure of DNA]

"Like my parents, I have never been a regular church member or churchgoer. It doesn't seem plausible to me that there is the kind of God who watches over human affairs, listens to prayers, and tries to guide people to follow His precepts – there is just too much misery and cruelty for that."

[Benjamin Spock]

"As long as men are free to ask what they must, free to say what they think, free to think what they will, freedom can never be lost, and science can never regress."

[J. Robert Oppenheimer, Life, 10 October 1949]

"There must be no barriers to freedom of inquiry. There is no place for dogma in science. The scientist is free, and must be free to ask any question, to doubt any assertion, to seek for any evidence, to correct any errors."

[J. Robert Oppenheimer, Life, 10 October 1949]

"On the ordinary view of each species having been independently created, we gain no scientific explanation..."

[Charles Darwin]

"Ignorance more frequently begets confidence than does knowledge: it is those who know little, and not those who know much, who so positively assert that this or that problem will never be solved by science."

[Charles Darwin, Introduction to "The Descent of Man, 1871]

"I can hardly see how anyone ought to wish Christianity to be true; for if so the plain language of the text seems to show that the men who do not believe, and this would include my Father, Brother and almost all my best friends, will be everlastingly punished. And this is a damnable doctrine."

[Charles Darwin]

"For myself, I do not believe in any revelation. As for a future life, every man must judge for himself between conflicting vague probabilities."

[Charles Darwin]

"They know that it is human nature to take up causes whereby a man may oppress his neighbor, no matter how unjustly. ... Hence they have had no trouble in finding men who would preach the damnability and heresy of the new doctrine from the very pulpit..."

[Galileo Galilei, 1615]

"I do not feel obliged to believe that same God who endowed us with sense, reason, and intel-

lect had intended for us to forgo their use."

[Galileo]

"It is surely harmful to souls to make it a heresy to believe what is proved."

[Galileo Galilei, "The Authority of Scripture in Philosophical Controversies"]

"To command the professors of astronomy to confute their own observations is to enjoin an impossibility, for it is to command them not to see what they do see, and not to understand what they do understand, and to find what they do not discover."

[Galileo Galilei, "The Authority of scripture in Philosophical Controversies"]

"It vexes me when they would constrain science by the authority of the Scriptures, and yet do not consider themselves bound to answer reason and experiment."

[Galileo Galilei, "The Authority of Scripture in Philosophical Controversies"]

"The church says the earth is flat, but I know that it is round, for I have seen the shadow on the moon, and I have more faith in a shadow than in the church."

[Ferdinand Magellan]

"It is our responsibility as scientists, knowing the great progress which comes from a satisfactory philosophy of ignorance, the great progress which is the fruit of freedom of thought, to proclaim the value of this freedom; to teach how doubt is not to be feared but welcomed and discussed; and to demand this freedom as our duty to all coming generations."

[Richard Feynman, "What Do You Care What Other People Think?"]

"The argument that the literal story of Genesis can qualify as science collapses on three major grounds: the creationists' need to invoke miracles in order to compress the events of the earth's history into the biblical span of a few thousand years; their unwillingness to abandon claims clearly disproved, including the assertion that all fossils are products of Noah's flood; and their reliance upon distortion, misquote, half-quote, and citation out of context to characterize the ideas of their opponents."

[Stephen Jay Gould, "The Verdict on Creation-

ism”,

The Skeptical Inquirer, Winter 87/88, pg. 186]

“In science, “fact” can only mean “confirmed to such a degree that it would be perverse to withhold provisional assent.” I suppose that apples might start to rise tomorrow, but the possibility does not merit equal time in physics classrooms.”

[Stephen J. Gould]

“The most important scientific revolutions all include, as their only common feature, the dethronement of human arrogance from one pedestal after another of previous convictions about our centrality in the cosmos.”

[Stephen Jay Gould, “Dinosaur in a Haystack”]

“Creation science” has not entered the curriculum for a reason so simple and so basic that we often forget to mention it: because it is false, and because good teachers understand exactly why it is false. What could be more destructive of that most fragile yet most precious commodity in our entire intellectual heritage – good teaching – than a bill forcing honorable teachers to sully their sacred trust by granting equal treatment to a doctrine not only known to be false, but calculated to undermine any general understanding of science as an enterprise?”

[Stephen Jay Gould, “The Skeptical Inquirer”]

“Our creationist detractors charge that evolution is an unproved and unprovable charade— a secular religion masquerading as science. They claim, above all, that evolution generates no predictions, never exposes itself to test, and therefore stands as dogma rather than disprovable science. This claim is nonsense. We make and test risky predictions all the time; our success is not dogma, but a highly probable indication of evolution’s basic truth.”

[Stephen Jay Gould, “Dinosaur in a Haystack”]

“I am not very skeptical... a good deal of skepticism in a scientific man is advisable to avoid much loss of time, but I have met not a few men, who... have often thus been deterred from experiments or observations which would have proven servicable.” - Charles Darwin

“Round about the accredited and orderly facts of every science there ever floats a sort of dust-cloud of exceptional observations, of occurrences minute and irregular and seldom

met with, which it always proves more easy to ignore than to attend to... Anyone will renovate his science who will steadily look after the irregular phenomena, and when science is renewed, its new formulas often have more of the voice of the exceptions in them than of what were supposed to be the rules.” - William James

“I know that most men, including those at ease with problems of the greatest complexity, can seldom accept even the simplest and most obvious truth if it be such as would oblige them to admit the falsity of conclusions which they have delighted in explaining to colleagues, which they have proudly taught to others, and which they have woven, thread by thread, into the fabric of their lives.” -Tolstoy

“It is really quite amazing by what margins competent but conservative scientists and engineers can miss the mark, when they start with the preconceived idea that what they are investigating is impossible. When this happens, the most well-informed men become blinded by their prejudices and are unable to see what lies directly ahead of them.” - Arthur C. Clarke, 1963

“When even the brightest mind in our world has been trained up from childhood in a superstition of any kind, it will never be possible for that mind, in its maturity, to examine sincerely, dispassionately, and conscientiously any evidence or any circumstance which shall seem to cast a doubt upon the validity of that superstition. I doubt if I could do it myself.” - Mark Twain

“Doubt everything or believe everything: these are two equally convenient strategies. With either we dispense with the need for reflection.” - Henri Poincare

“It is not uncommon for engineers to accept the reality of phenomena that are not yet understood, as it is very common for physicists to disbelieve the reality of phenomena that seem to contradict contemporary beliefs of physics” - H. Bauer

“There is a principle which is a bar against all information, which cannot fail to keep a man in everlasting ignorance— that principle is contempt prior to investigation.” - Herbert Spencer, British philosopher

“If a man is in too big a hurry to give up an error he is liable to give up some truth with it.” - Wilbur Wright, 1902

"It's like religion. Heresy [in science] is thought of as a bad thing, whereas it should be just the opposite." - Dr. Thomas Gold

"You can get into a habit of thought in which you enjoy making fun of all those other people who don't see things as clearly as you do. We have to guard carefully against it." - Carl Sagan, 1987 CSICOP meeting

"New and stirring things are belittled because if they are not belittled, the humiliating question arises, 'Why then are you not taking part in them?' " - H. G. Wells

"The easy confidence with which I know another man's religion is folly teaches me to suspect that my own is also." - Mark Twain

"I believe there is no source of deception in the investigation of nature which can compare with a fixed belief that certain kinds of phenomena are IMPOSSIBLE." -William James

"Modern science should indeed arouse in all of us a humility before the immensity of the unexplored and a tolerance for crazy hypotheses." -Martin Gardner

"Almost all really new ideas have a certain aspect of foolishness when they are first produced." - Alfred North Whitehead

"The mind likes a strange idea as little as the body likes a strange protein and resists it with similar energy. It would not perhaps be too fanciful to say that a new idea is the most quickly acting antigen known to science." - Wilfred Trotter, 1941

"When a distinguished but elderly scientist states that something is possible, he is almost certainly right. When he states that something is impossible, he is very probably wrong." - Arthur C. Clarke's First Law

"There are some people that if they don't know, you can't tell 'em." - Louis Armstrong

"The security provided by a long-held belief system, even when poorly founded, is a strong impediment to progress. General acceptance of a practice becomes the proof of its validity, though it lacks all other merit." - Dr. B. Lown, invented defibrillator

"The fact that an opinion has been widely held is no evidence whatever that it is not utterly absurd; indeed in view of the silliness of the majority of mankind, a widespread belief is

more likely to be foolish than sensible." - Bertrand Russell

"New opinions are always suspected, and usually opposed, without any other reason but because they are not already common." - John Locke

"All great truths begin as blasphemies." - George Bernard Shaw

"Be not astonished at new ideas; for it is well known to you that a thing does not therefore cease to be true because it is not accepted by many." - Spinoza

"If we watch ourselves honestly we shall often find that we have begun to argue against a new idea even before it has been completely stated." - Wilfred Trotter

"When a man finds a conclusion agreeable, he accepts it without argument, but when he finds it disagreeable, he will bring against it all the forces of logic and reason." -Thucydides

"It is difficult to say what is impossible, for the dream of yesterday is the hope of today and the reality of tomorrow." - Robert Goddard

"Science might be better served when some scientists generate novel ideas while others carp at everything new, than if all scientists could somehow become disinterestedly skeptical." Dr. Henry H. Bauer

"Type one' error is thinking that something special is happening when nothing special really is happening. 'Type two' error is thinking that nothing special is happening, when in fact something rare or infrequent is happening.' -M. Truzzi

"I ask you, which is the greater threat to science and mankind, accepting a claim that can have no possible benefit, or rejecting a claim that can have great benefit?" -Dr. Edmund Storms

"There is nothing particularly scientific about excessive caution. Science thrives on daring generalizations." - L. Hogben

"What we need is not the will to believe but the will to find out." - Bertrand Russell

"If you haven't found something strange during the day, it hasn't been much of a day." - J. A. Wheeler

"Only those who attempt the absurd will achieve the impossible." - M. C. Escher

"What is there that confers the noblest delight? What is that which swells a man's breast with pride above that which any other experience can bring to him? Discovery! To know that you are walking where none others have walked..." - Mark Twain

"Man's greatest asset is the unsettled mind." - Isaac Asimov

"It would seem to me... an offense against nature, for us to come on the same scene endowed as we are with the curiosity, filled to overbrimming as we are with questions, and naturally talented as we are for the asking of clear questions, and then for us to do nothing about, or worse, to try to suppress the questions..." - Lewis Thomas

"Only a fool of a scientist would dismiss the evidence and reports in front of him and substitute his own beliefs in their place." - Paul Kurtz

"The creative person pays close attention to what appears discordant and contradictory... and is challenged by such irregularities." - F. Barron

"Genius in truth means little more than the faculty of perceiving in an unhabitual way" - William James, 1896

"The voyage of discovery lies not in seeking new horizons, but in seeing with new eyes." - Marcel Proust

"Research is to see what everybody else has seen, and to think what nobody else has thought." - Albert Szent-Gyorgi

"A man receives only what he is ready to receive... The phenomenon or fact that cannot in any wise be linked with the rest of what he has observed, he does not observe." - H. D. Thoreau

"You cannot depend on your eyes when your imagination is out of focus." - Mark Twain

"The man who cannot occasionally imagine events and conditions of existence that are contrary to the causal principle as he knows it will never enrich his science by the addition of a new idea." - Max Planck

"We not only believe what we see, to some extent we see what we believe ...The implica-

tions of our beliefs are frightening." - Richard Gregory

"If what we regard as real depends on our theory, how can we make reality the basis of our philosophy? ...But we cannot distinguish what is real about the universe without a theory...it makes no sense to ask if it corresponds to reality, because we do not know what reality is independent of a theory." - Stephen Hawking

"If we knew what it was we were doing, it would not be called research, would it?" - Albert Einstein

"Exploratory research is really like working in a fog. You don't know where you're going. You're just groping. Then people learn about it afterwards and think how straightforward it was." - Francis Crick

"Reality is that which, when you stop believing in it, doesn't go away." - Phillip K. Dick

"There are children playing in the street who could solve some of my top problems in physics, because they have modes of sensory perception that I lost long ago." - Robert Oppenheimer

"The discovery of truth is prevented more effectively not by the false appearance of things present and which mislead into error, not directly by weakness of the reasoning powers, but by preconceived opinion, by prejudice." - Schopenhauer

"It is a puzzling thing. The truth knocks on the door and you say, 'Go away, I'm looking for the truth.' and so it goes away. Puzzling." - R. Pirsig

"They are ill discoverers that think there is no land when they see nothing but sea." - Francis Bacon

"The universe is wider than our views of it." - Henry David Thoreau

"Everyone takes the limits of his own vision for the limits of the world." - Arthur Schopenhauer

"Who never walks save where he sees men's tracks makes no discoveries." - J.G. Holland

"In questions of science, the authority of a thousand is not worth the humble reasoning of a single individual." - Galileo Galilei

"It is as fatal as it is cowardly to blink facts because they are not to our taste." - John Tyndall

"Many discoveries must have been stillborn or smothered at birth. We know only those which survived." - W. I. Beveridge, THE ART OF SCI. INVESTIGATION

"A foolish consistency is the hobgoblin of small minds." - Emerson

"Absence of evidence is not evidence of absence." - Sir Martin Rees (astronomer)

"I can't see any farther. Giants are standing on my shoulders!" - unknown

"In science it often happens that scientists say, 'You know that's a really good argument; my position is mistaken,' and then they would actually change their minds and you never hear that old view from them again. They really do it. It doesn't happen as often as it should, because scientists are human and change is sometimes painful. But it happens every day. I cannot recall the last time something like that happened in politics or religion." -Carl Sagan

"When I examined myself and my methods of thought, I came to the conclusion that the gift of fantasy has meant more to me than my talent for absorbing positive knowledge." - A. Einstein

"All truths are easy to understand once they are discovered; the point is to discover them." - Galileo Galilei

"Advances are made by answering questions. Discoveries are made by questioning answers." - Bernhard Haisch, astrophysicist

"The most erroneous stories are those we think we know best – and therefore never scrutinize or question." -Stephen Jay Gould

"It is a good morning exercise for a research scientist to discard a pet hypothesis every day before breakfast. It keeps him young." - Konrad Lorenz

"Inquiry is fatal to certainty." - William J. Durant

"In every work of genius we recognize our own rejected thoughts." - Ralph Waldo Emerson

"There is no better soporific and sedative than skepticism." -Nietzsche

"...By far the most usual way of handling phenomena so novel that they would make for a serious rearrangement of our preconceptions is to ignore them altogether, or to abuse those

who bear witness for them." - William James

"Science today is locked into paradigms. Every avenue is blocked by beliefs that are wrong, and if you try to get anything published by a journal today, you will run against a paradigm and the editors will turn it down" - Sir Fred Hoyle

"If we will only allow that, as we progress, we remain unsure, we will leave opportunities for alternatives. We will not become enthusiastic for the fact, the knowledge, the absolute truth of the day, but remain always uncertain... In order to make progress, one must leave the door to the unknown ajar." - Richard Feynman

"The pressure for conformity is enormous. I have experienced it in editors rejection of submitted papers, based on venomous criticism of anonymous referees. The replacement of impartial reviewing by censorship will be the death of science." -Julian Schwinger, physicist

"When adults first become conscious of something new, they usually either attack or try to escape from it... Attack includes such mild forms as ridicule, and escape includes merely putting out of mind." - W. I. B Beveridge, The Art of Sci. Investigation, 1950

"New ideas are always criticized - not because an idea lacks merit, but because it might turn out to be workable, which would threaten the reputations of many people whose opinions conflict with it. Some people may even lose their jobs." - physicist, requested anonymity

"Too much openness and you accept every notion, idea, and hypothesis - which is tantamount to knowing nothing. Too much skepticism - especially rejection of new ideas before they are adequately tested - and you're not only unpleasantly grumpy, but also closed to the advance of science. A judicious mix is what we need." - Carl Sagan

"All truth passes through three stages: First, it is ridiculed; Second, it is violently opposed; and Third, it is accepted as self-evident." - Arthur Schopenhauer

"Theories have four stages of acceptance: i) this is worthless nonsense; ii) this is an interesting, but perverse, point of view; iii) this is true, but quite unimportant; iv) I always said so. -J.B.S. Haldane, 1963

"When a thing is new, people say: 'It is not true.' Later, when its truth becomes obvious, they

say: 'It is not important.' Finally, when its importance cannot be denied, they say: 'Anyway, it is not new.'" - William James, 1896

"The radical invents the views. When he has worn them out the conservative adopts them." - Mark Twain

"To imagine that turmoil is in the past and somehow we are now in a more stable time seems to be a psychological need." - geologist E. Moores

"The soft-minded man always fears change. He feels security in the status quo, and he has an almost morbid fear of the new. For him, the greatest pain is the pain of a new idea." - Dr. Martin Luther King Jr.

"Loyalty to a petrified opinion never yet broke a chain or freed a human soul." - Mark Twain

"No Pessimist ever discovered the secrets of the stars, or sailed to an uncharted land, or opened a new heaven to the human spirit" - Helen Keller

"A danger sign of the lapse from true skepticism in to dogmatism is an inability to respect those who disagree" - Dr. Leonard George

"We should be eternally vigilant against attempts to check the expression of opinions that we loathe." - Oliver Wendell Holmes

"If I want to stop a research program I can always do it by getting a few experts to sit in on the subject, because they know right away that it was a fool thing to try in the first place." - Charles Kettering, GM

"If you are only skeptical, then no new ideas make it through to you. You become a crotchety old person convinced that nonsense is ruling the world. (There is, of course, much data to support you.) But every now and then, a new idea turns out to be on the mark, valid and wonderful. If you are too much in the habit of being skeptical about everything, you are going to miss or resent it, and either way you will be standing in the way of understanding and progress." - Carl Sagan

"There is a very important distinction between a critical attitude of mind (or critical "faculty") and a sceptical attitude." - W. Beveridge

"In philosophical discussion, the merest hint of dogmatic certainty as to finality of statement is an exhibition of folly." - Whitehead

"There is a principle which is a bar against all information, which cannot fail to keep a man in everlasting ignorance—that principle is contempt prior to investigation." - Herbert Spencer, British philosopher

"It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts." - Sherlock Holmes (Sir Arthur Conan Doyle)

"Now, my suspicion is that the universe is not only queerer than we suppose, but queerer than we can suppose... I suspect that there are more things in heaven and earth than are dreamed of, in any philosophy" - J.B.S. Haldane

"The farther the experiment is from theory, the closer it is to the Nobel Prize." - Joliet-Curie

"There are two possible outcomes: If the result confirms the hypothesis, then you've made a measurement. If the result is contrary to the hypothesis, then you've made a discovery." - Enrico Fermi

"Daring ideas are like chessmen moved forward; they may be defeated, but they start a winning game." - Goethe

"Everything we know is only some kind of approximation, because we know that we do not know all the laws yet. Therefore, things must be learned only to be unlearned again or, more likely, to be corrected." - Richard Feynman

"As long as we do science, some things will always remain unexplained." - Fritjof Capra

"The philosophies of one age have become the absurdities of the next, and the foolishness of yesterday has become the wisdom of tomorrow." - Sir William Osler

"The altar cloth of one aeon is the doormat of the next." - Mark Twain

"Perhaps the only thing that saves science from invalid conventional wisdom that becomes effectively permanent is the presence of mavericks in every generation - people who keep challenging convention and thinking up new ideas for the sheer hell of it or from an innate contrariness." - Dr. D. M. Raup, Paleontologist, U. Chicago.

"One thing I have learned in a long life: that all our science, measured against reality, is primitive and childlike - and yet it is the most precious thing we have." - Einstein

"We do not understand much of anything, from... the "big bang", all the way down to the particles in the atoms of a bacterial cell. We have a wilderness of mystery to make our way through in the centuries ahead." -Lewis Thomas

"There is no natural phenomenon that is comparable with the sudden and apparently accidentally timed development of science, except perhaps the condensation of a super-saturated gas or the explosion of some unpredictable explosives." - Eugene P. Wigner

"Beware when the great God lets loose a thinker on this planet. Then all things are at risk. It is as when a conflagration has broken out in a great city, and no man knows what is safe, or where it will end." - Ralph Waldo Emerson

"Fiction is obliged to stick to possibilities. Truth isn't." - Mark Twain

"Nothing is too wonderful to be true if it be consistent with the laws of nature." - Michael Faraday

"The love of the marvelous is the most dangerous enemy of natural science." - mineralogist Eugene de Patrin, said in 1802 while dismissing reports of meteorites.

"The skeptic will say, 'It may well be true that this system of equations is reasonable from a logical standpoint, but this does not prove that it corresponds to nature.' You are right, dear skeptic. Experience alone can decide on truth." - Albert Einstein

Unnamed Law: If it happens, it must be possible.

"I have steadily endeavored to keep my mind free so as to give up any hypothesis, however much beloved (and I cannot resist forming one on every subject), as soon as the facts are shown to be opposed to it." - Charles Darwin

"I love fools' experiments, I am always making them." - Darwin

"It is a fool's prerogative to utter truths that no one else will speak." - Shakespeare

"The whole of science consists of data that, at one time or another, were inexplicable." - B. O'Regan

"Name the greatest of all the inventors. Accident." -Mark Twain

"May every young scientist remember... and not fail to keep his eyes open for the possibility, that an irritating failure of his apparatus to give consistent results may once or twice in a lifetime conceal an important discovery." - P. Blackett

"My advice to those who wish to learn the art of scientific prophesy is not to rely on abstract reason, but to decipher the secret language of Nature from Nature's documents: the facts of experience." - Max Born

"The beginning of knowledge is the discovery of something we do not understand." - Frank Herbert

"In any field, find the strangest thing and then explore it." - John A. Wheeler

"The most exciting phrase to hear in science, the one that heralds new discoveries, is not 'Eureka!' (I found it!) but 'That's funny...'" - Isaac Asimov

"The only solid piece of scientific truth about which I feel totally confident is that we are profoundly ignorant about nature... It is this sudden confrontation with the depth and scope of ignorance that represents the most significant contribution of twentieth-century science to the human intellect." - Lewis Thomas

"The end of our exploring will be to arrive at where we started, and to know the place for the first time." - T.S. Eliot

"Sit down before facts like a child, and be prepared to give up every preconceived notion, follow humbly wherever and to whatever abysses Nature leads, or you shall learn nothing." - T.H. Huxley

"Truth is stranger than fiction, but it is because Fiction is obliged to stick to possibilities; Truth isn't." - Mark Twain

"Let the mind be enlarged... to the grandeur of the mysteries, and not the mysteries contracted to the narrowness of the mind" - Francis Bacon

"Man's mind stretched to a new idea never goes back to its original dimension." - Oliver Wendell Holmes

"The test of a first-rate intelligence is the ability to hold two opposed ideas in mind at the same time and still retain the ability to function." - F. Scott Fitzgerald

"It is the mark of an educated mind to be able to entertain a thought without accepting it."

-Aristotle

"I can live with doubt and uncertainty and not knowing. I think it is much more interesting to live not knowing than to have answers that might be wrong." - Richard Feynman

"You cannot teach a man anything, you can only help him find it within himself." - Galileo

"The high-minded man must care more for the truth than for what people think." - Aristotle

"In real life, every field of science is incomplete, and most of them - whatever the record of accomplishment during the last 200 years - are still in their very earliest stages." - Lewis Thomas

"There are many hypotheses in science which are wrong. That's perfectly all right; they're the aperture to finding out what's right." - Carl Sagan

"I personally feel it is presumptuous to believe that man can determine the whole temporal structure of the universe, its evolution, development and ultimate fate from the first nanosecond of creation to the last 10^{10} years, on the basis of three or four facts which are not very accurately known and are disputed among the experts." - J. Bahcall, senior astrophysicist, Institute for Advanced Study

"On any Tuesday morning, if asked, a good working scientist will tell you with some self-satisfaction that the affairs of his field are nicely in order, that things are finally looking clear and making sense, and all is well. But come back again on another Tuesday, and the roof may have just fallen in on his life's work." -Lewis Thomas

"Science for me is very close to art. Scientific discovery is an irrational act. It's an intuition which turns out to be reality at the end of it -and I see no difference between a scientist developing a marvelous discovery and an artist making a painting." - C. Rubbia, Nobelist and director of CERN

"It is through science that we prove, but through intuition that we discover." - H. Poincare

"Science... is part and parcel of our knowledge and obscures our insight only when it holds that the understanding given by it is the only kind there is." - C.G. Jung

"The person who thinks there can be any real conflict between science and religion must be either very young in science or very ignorant of religion." - Joseph Henry, early American physicist

"Science is not only compatible with spirituality; it is a profound source of spirituality." - Carl Sagan

"Science without religion is lame. Religion without science is blind." -Albert Einstein

"The intuitive mind is a sacred gift and the rational mind is a faithful servant. We have created a society that honors the servant and has forgotten the gift." - Albert Einstein

"If you restrict the journal to publishing only what pleases the referees, you end up publishing what is popular, and while it does make everyone feel more comfortable, you are guaranteed to miss the occasional breakthrough." - A. Dessler, Editor, Geophysical Research Letters, (regarding small-comet bombardment of Earth.)

"No matter how we may single out a complex from nature...its theoretical treatment will never prove to be ultimately conclusive... I believe that this process of deepening of theory has no limits." - Albert Einstein, 1917

"Biologists can be just as sensitive to heresy as theologians." - H.G. Wells

"A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die and a new generation grows up that is familiar with it." - M. Planck

"Science advances funeral by funeral."-Planck

"When the human race has once acquired a superstition, nothing short of death is ever likely to remove it." - Mark Twain

"You can recognize a pioneer by the arrows in his back." - Beverly Rubik

"If the man doesn't believe as we do, we say he is a crank, and that settles it. I mean, it does nowadays, because now we can't burn him." - Mark Twain

"Scientists are not the paragons of rationality, objectivity, openmindedness and humility that many of them might like others to believe." - Marcello Truzzi, CSICOP

"The common idea that scientists reject a theory as soon as it leads to a contradiction is just not so. When they get something that works at all they plunge ahead with it and ignore its weak spots... scientists are just as bad as the rest of the public in following fads and being influenced by mass enthusiasm." - Vannevar Bush

"Once a new paradigm takes hold, its acceptance is extraordinarily rapid and one finds few who claim to have adhered to a discarded method." - Dr. B. Lown, inventor of the modern defibrillator

"For every expert, there is an equal and opposite expert." - anon

"One could not be a successful scientist without realizing that, in contrast to the popular conception supported by newspapers and mothers of scientists, a goodly number of scientists are not only narrow-minded and dull, but also just stupid." - J. D. Watson "The Double Helix"

As a whole, parapsychologists are nice, honest people, while the critics are cynical, nasty people" - Ray Hyman, skeptical scientist, 1985

"Desire for approval and recognition is a healthy motive, but the desire to be acknowledged as better, stronger, or more intelligent than a fellow being or fellow scholar easily leads to an excessively egoistic psychological adjustment, which may become injurious for the individual and for the community." - Albert Einstein

"Science is the search for truth - it is not a game in which one tries to beat his opponent, to do harm to others." - Linus Pauling

"The need to be right all the time is the biggest bar to new ideas. It is better to have enough ideas for some of them to be wrong, than to be always right by having no ideas at all." - Edward de Bono

"To swear off making mistakes is very easy. All you have to do is swear off having ideas." - Leo Burnett

"A man with a new idea is a crank until he succeeds." - M. Twain

"Don't worry about people stealing your ideas. If your ideas are that good, you'll have to ram them down people's throats." - Howard Aiken

"Physical concepts are the free creations of the human mind and are not, however it may seem, uniquely determined by the external world." - Einstein/Infeld in "The Evolution of Physics" 1938

"A new idea is delicate. It can be killed by a sneer or a yawn; it can be stabbed to death by a joke, or worried to death by a frown on the right person's brow." - Charles Brower

"A great many people think they are thinking when they are merely rearranging their prejudices." - William James

"If you make people think they're thinking, they'll love you; but if you really make them think they'll hate you." - Don Marquis

"We must care to think about the unthinkable things, because when things become unthinkable, thinking stops and action becomes mindless." - James W. Fulbright

"Wisest is she who knows she does not know." -anon

"The only means of strengthening one's intellect is to make up one's mind about nothing - to let the mind be a thoroughfare for all thoughts. Not a select party." - John Keats

"There is nothing so absurd that it cannot be believed as truth if repeated often enough." -William James

"A lie repeated often enough becomes the truth." - G. Goebbels

"Never attribute to conspiracy that which is adequately explained by stupidity." - paraphrase of "Hanlon's Razor" (fm R. Heinlein)

What I don't understand I despise, what I despise I reject. - THE REFEREE'S CREED

"Without deviation from the norm, progress is not possible." - Frank Zappa

"My friends they were dancing here in the streets of Huntsville when our first satellite orbited the Earth. They were dancing again when the first Americans landed on the moon. I'd like to ask you, don't hang up your dancing slippers." - Wernher von Braun

The greatest enemy of knowledge is not ignorance, it is the illusion of knowledge.

Stephen Hawking